

POST-TREATMENT GRANULOMATOUS INFLAMMATION MIMICKING LOCAL BREAST CANCER RECURRENCE:

THE VALUE OF MULTIDISCIPLINARY RADIOLOGIC–PATHOLOGIC CORRELATION

Reihana Zahra, Windu Cahyaningrum Handayani Notonagoro Suryaningrat, Rahmi Afifi, Sawitri Darmiati

Radiology Residency Program, Faculty of Medicine Universitas Indonesia - Dr. Cipto Mangunkusumo National Referral Hospital, Jakarta, Indonesia • Division of Woman's Imaging, Department of Radiology, Faculty of Medicine Universitas Indonesia - Dr. Cipto Mangunkusumo National Referral Hospital, Jakarta, Indonesia • Department of Radiology, Faculty of Medicine Universitas Indonesia, Jakarta, Indonesia



BACKGROUND

Post-treatment breast abnormalities may closely mimic local recurrence, creating a diagnostic and management challenge in breast cancer survivors. **Granulomatous inflammatory reactions** are uncommon benign mimics that **may demonstrate suspicious clinical and imaging features**. Accurate distinction requires integration of imaging, pathology, treatment history, and longitudinal assessment.

CASE PRESENTATION

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- Woman with right breast cancer
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- Underwent **BCS**, followed by **adjuvant** and **endocrine** therapy
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- On surveillance, developed **erythema and firmness** in treated breast
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- 4.1-cm ill-defined heterogeneous irregular lesion in the superomedial right breast.
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- Irregular **hypoechoic lesion** with posterior shadowing, perilesional vascularity, and edema, **without suspicious regional lymphadenopathy**
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- Planned **mastectomy** considered
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- Core biopsy for confirmation**
Chronic foreign-body granulomatous inflammation; no residual tumor

TAKE-HOME MESSAGE

Suspicious does not always mean recurrent: confirm with pathology and follow the lesion over time before escalating treatment.

IMAGING FINDINGS

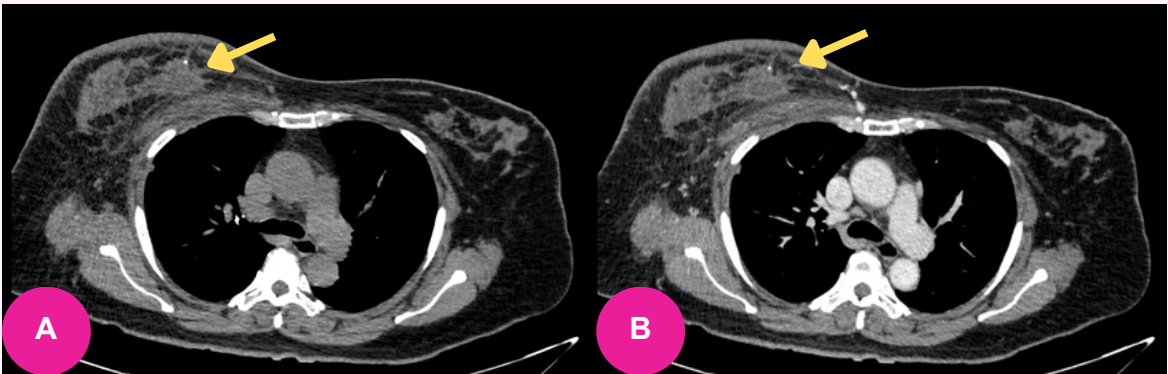


Figure A dan B. Axial Contrast-enhanced thoracoabdominal CT demonstrated a 4.1-cm ill-defined heterogeneous irregular enhancing lesion in the superomedial right breast, suspicious for local recurrence.

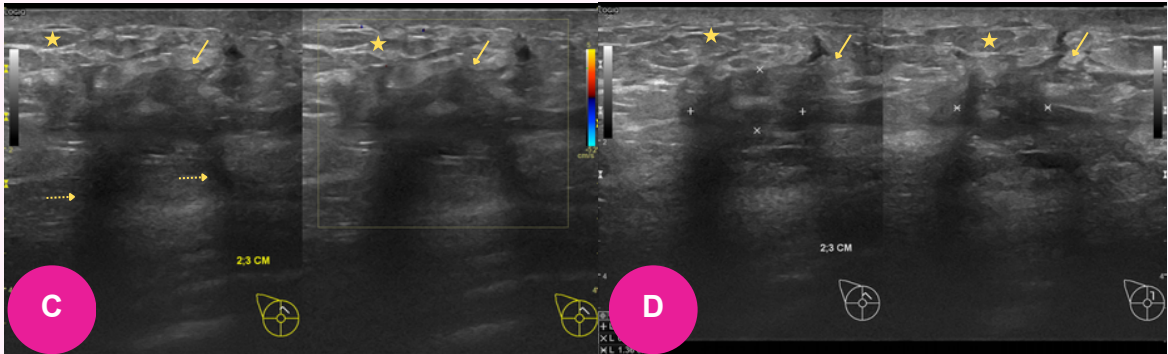


Figure C dan D. Targeted breast ultrasound shows an irregular heterogeneous hypoechoic lesion (arrow) with posterior shadowing (dash), perilesional vascularity, and surrounding skin and subcutaneous edema (star).

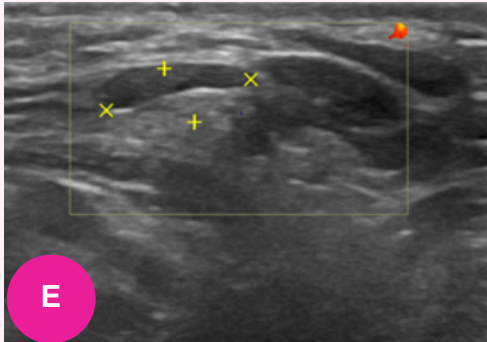


Figure E. Ultrasound of the right axillary tail showed an axillary lymph node with a preserved fatty hilum and no cortical thickening, consistent with benign nodal morphology.

DISCUSSION

- Suspicious breast findings detected on CT should undergo dedicated diagnostic mammography/DBT with targeted ultrasonography; MRI may be reserved for problem-solving when conventional imaging remains inconclusive.
- Mammography/DBT: Best for detecting new calcifications, architectural distortion, and asymmetry with comparison to prior tumor-bed imaging.
- Targeted US: Best for localized lesion characterization, inflammatory changes, nodal assessment, and biopsy guidance.
- Contrast-enhanced MRI: Best as a problem-solving tool for equivocal findings and for assessing disease extent.
- Features** to help evaluate **likelihood of recurrence**:

Original cancer biology	Imaging Behaviour	Concordance
Higher concern: TNBC/HER2+, high grade, LVI+, node+, large tumor, positive margin, younger age	New or enlarging (progressive) focal abnormality favors recurrence. Post treatment changes usually shows inflammatory change that decreases	Discordant: suspicious imaging → biopsy shows inflammation → lesion persists or enlarges. Recurrence cannot be excluded ; repeat biopsy or further evaluation is warranted.

- Original tumor location, surgical bed, margins, and pathology help distinguish recurrence from post-treatment change.
- Temporal evolution and radiologic–pathologic concordance should guide management.

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